AQUACULTURE BIOSECURITY





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INTRODUCTION

❑WHAT IS BIOSECURITY?

❖The term biosecurity first referred to biological weapons and bio-terrorism.

❖Originally used in reference to the management of biological weapons and bio-terrorism, the term biosecurity was first used in the **agricultural sector** in the 1980s as **"the sum of risk management practices in the defence against biological threats"**

Biosecurity is the exclusion, eradication or effective management of risks posed by pests and diseases to the economy, environment and human health.

INTRODUCTION

❑WHAT IS BIOSECURITY?

❖Defined as “a set of management and physical measures designed to mitigate the risk of introduction of pathogenic agents into, or spread within, or release from, aquatic animal production”.

INTRODUCTION

❑WHAT IS BIOSECURITY?

- ❖It is now used in diverse sectors including biological laboratory risks and infectious disease prevention in both animal and public health.
- ❖Health management in the aquaculture sector must be applied and addressed for sustainability purposes.

INTRODUCTION

- The definition of the biosecurity concept by the Food and Agriculture Organization (FAO) is a strategic and integrated approach that encompasses the policy and regulatory frameworks for analyzing and managing relevant risks to human, animal, and plant life and health and the environment (FAO 2007, 2020).
- The concept is applied to production sectors like; Agriculture, poultry, animal husbandry, and aquaculture industries, which need proper health management of their production system.

INTRODUCTION

- The rapid growth of the global aquaculture industry has led to a dramatic increase in the spread of pathogens (e.g., pests, bacteria, viruses, and other parasites) by translocation of commercial aquatic organisms, such as fishes, shrimp, and mollusks
- These pathogens causes economic loss, reduction in profit and revenues and poor food security systems, low market accessibility.
- Preventing and controlling pathogen spread is a major concern of aquaculture biosecurity starting from on-farm to national and international levels, particularly when unrecognized species enter transport pathways.
- As "prevention is better than cure," it is advisable to focus on preventing the occurrence of pathogens in the aquaculture system.

PROTECT YOUR FARM IN SEVEN STEPS

1.Maintain stock health and welfare

- Follow good animal husbandry practices.
- The benefits of good animal welfare and biosecurity go hand-in-hand and should minimise animal stress where practicable.

2.Be aware of the pathways of risk organisms onto, within and off your farm

- Make sure your staff are familiar with the pests and diseases of concern and their pathways.
- Carry out biosecurity induction and ongoing training to explain required behaviours and hygiene practices on your farm.

PROTECT YOUR FARM IN SEVEN STEPS

3.Obtain pest and disease-free stock

- Preferentially source stock from areas/facilities with a known health status and with a health monitoring programme.
- Keep records of all stock movements onto, within and off your farm so they can be traced in the event of a problem.

4. Keep things clean

- Carry out good cleaning practices to help prevent the introduction, exacerbation and spread of pests and diseases.
- All people (including workers and visitors), equipment, vehicles and vessels represent potential ways of spreading pests or diseases onto, within or off your farm.

PROTECT YOUR FARM IN SEVEN STEPS

5.Check your farm

- Constant vigilance is vital for early detection of pests and diseases.
- Regularly monitor your stock for signs of unusual behaviour, colourations, pests or mortalities.
- Knowing what is normal will help you to recognise what isn’t.
- Keep records (written, photographic) of all unusual observations.

PROTECT YOUR FARM IN SEVEN STEPS

6.Report anything unusual

- Farmers are regularly in and around the aquatic environment and therefore are well placed to notice anything out of the ordinary.
- If your stock appears abnormal or experiences unusual sickness or mortalities.

7.Have a biosecurity management plan in place

- A biosecurity management plan will help you to assess and prioritise the biosecurity risks relevant to your farming operation.
- Implementing good biosecurity practices will in turn help you prevent, reduce or manage those risks.

MAJOR WAYS OF PATHOGEN TRANSMISSION

- Aquaculture biosecurity can be effectively managed when there is an understating of pathways that can introduce, take away or reinfect the farm/stock (incoming, outgoing or enhancing the infection).

- Examples of pathways include;

1. Stock movements and containment

✓Introduction of new stock or broodstock onto your farm.

✓Stock movements within your farm.

✓Stock movements off your farm.

✓Stock escapes.

MAJOR WAYS OF PATHOGEN TRANSMISSION

2. Water

- ✓Water that can be controlled -pumped or flowed onto, within and off your farm
- ✓Uncontrolled water – farms that reside in water e.g. in lakes
- ✓Water that stock is transported in.

3. Equipment

- ✓Nets, ropes, baskets, buoys, containers, clothing, footwear
- ✓Vessels – supply boats
- ✓Vehicles – transporting trucks, trailers, courier vans.

MAJOR WAYS OF PATHOGEN TRANSMISSION

4. People

✓Site staff

✓Staff from other farms

✓Visitors, contractors, couriers, suppliers.

5. Feed

✓Stock feed

✓Raw materials.

6. Wildlife

✓Marine mammals, birds

✓Vermin (rats, mice, stray or feral animals)

✓Domestic animals.

MAJOR WAYS OF PATHOGEN TRANSMISSION

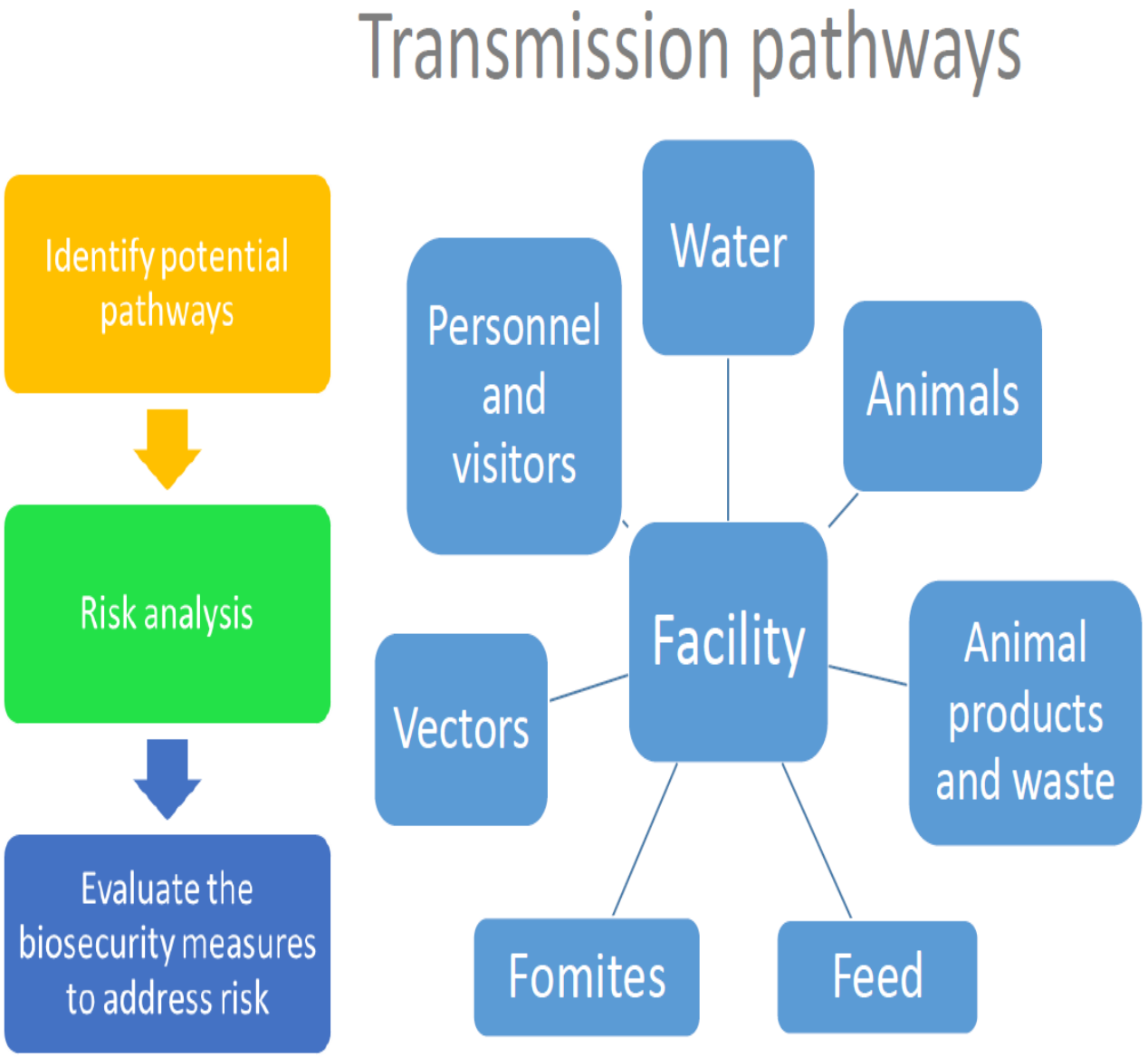




MAJOR WAYS OF PATHOGEN TRANSMISSION



TRANSMISSION PATHWAYS





COMING UP WITH A BIOSECURITY PLAN

- Developing a plan comes with having clearer biosecurity Management Objectives and simple practices.
- The following are some of the farm biosecurity plans and practices which can be executed.

Good practices should not be overly complicated, costly, or difficult to follow and implement.

COMING UP WITH A BIOSECURITY PLAN

❑1. STOCK HEALTH MANAGEMENT

✓**Objective:** Stock health should be maintained to optimum levels.

❖**Simple practices.**

➢Keep stock stress to a minimum.

➢Maintain optimum water quality levels.

➢Maintain appropriate stocking densities.

➢Provide appropriate diets and nutrition.

➢Avoid unnecessary or rough handling of stock.

➢Keep transfers within or between farms to a minimum.

➢Monitor stock frequently.

➢Remove dead or dying stock from remaining stock as soon as practicable.

➢Monitor and keep records of all aspects of stock health management

COMING UP WITH A BIOSECURITY PLAN

❑2. STOCK MOVEMENTS AND CONTAINMENT

✓Objective: Minimise the pest and disease risk associated with stock movements onto, within and off your farm.

❖Simple practices

➢Only stock of known health status should be introduced onto your farm.

➢Only move stock off your farm to locations of equal or lesser health status than the current location.

➢Prevent stock escapes.

➢If stock of unknown health status is to be introduced (e.g. wild broodstock, seed stock), this stock should be isolated in separate production units or dedicated quarantine facilities (e.g. land-based farms)

➢Precautions should be undertaken to prevent the within farm spread of pests, disease until such situations are resolved

➢Keep accurate records of all stock movements onto, within or off your farm.

COMING UP WITH A BIOSECURITY PLAN

❑3. Water sources (land-based farms)

✓Objective: Minimise the risks of pests and disease entry and spread associated with incoming water.

❖Simple practices

➢The water supply should be assessed for biosecurity risk and appropriate action(s) taken

➢Locate water intake and outflow pipes to avoid cross-contamination.

➢ Treat intake water to an acceptable level so as to minimise the risk of pests and diseases entering your farm (screening, filtration, ultra-violet light, ozone).

➢Install screens on intake pipes to prevent entry of undesirable organisms.

➢Maintain good water quality at all times

COMING UP WITH A BIOSECURITY PLAN

❑ 4. EQUIPMENT, VEHICLES AND VESSELS

✓Objective: Minimise the risk of equipment, vehicles and vessels introducing and spreading pests and diseases onto, within and off your farm.

❖Simple practices

- Assess all equipment, vehicles and vessels entering the farm for biosecurity risk and ensure that risk is managed by taking appropriate action(s)
- Farms should have dedicated and well-designed delivery and loading areas
- Standard operating procedures and dedicated infrastructure should be in place for cleaning and disinfection of equipment, vehicles and vessels

COMING UP WITH A BIOSECURITY PLAN

❑5. PEOPLE MANAGEMENT: Staff and visitors

❖Objective: Minimise the risk of staff and visitors transferring pests and diseases onto, within and off your farm.

✓**Simple practices**

➢Assess all staff and visitor access to farms for biosecurity risk and ensure that risk is managed by taking appropriate action(s).

➢Refusal of entry should be considered for high risk visitors

➢Manage farm access (e.g. access controls and signage)

➢All visitors should be briefed regarding on-farm biosecurity

➢Restrict staff and visitor access to sensitive areas.

➢Keep records of all visitors to your farm and education and staff training

COMING UP WITH A BIOSECURITY PLAN



Figure 1. Mobile footbath made from a shallow container.





Figure 6. A drive-through disinfection area, consisting of a shallow area filled with disinfectant, to disinfect vehicle wheels.



COMING UP WITH A BIOSECURITY PLAN

❑ 6. FEEDS AND FEEDING

✓Objective: Minimise the risk of feed exacerbating or transferring pests and diseases onto, within and off your farm.

❖Simple practices

➢All feeds entering the farm should be assessed for biosecurity risk and appropriate action(s) taken to address identified risk(s)

➢Source feeds from reputable suppliers to ensure they provide assurances of quality and content

➢Ensure any feed containing aquatic organisms is adequately treated to ensure safe product

➢Store feeds in designated areas (e.g. clean and dry) to avoid contamination and reduced feed quality

➢Regularly inspect feed to check for the presence of mould, vermin, and other undesirable organisms.

COMING UP WITH A BIOSECURITY PLAN

❑7. WILDLIFE, SCAVENGERS AND VERMIN

✓Objective: Minimise the risk of wildlife, scavengers, vermin, and pets transferring pests and disease onto, within or from the farm.

❖Simple practices

➢Control or exclude predators, wildlife, vermin and other organisms (e.g. aquatic life) from land-based systems

➢Control, exclude or prevent aggregations of predators, wildlife, vermin and other organisms (e.g. aquatic life) from open water systems

➢Regularly inspect farms for biosecurity breaches or signs of potential breaches and remedy as required

➢Keep records of any presence of wildlife and vermin, or biosecurity breaches and any preventive or corrective actions taken.

COMING UP WITH A BIOSECURITY PLAN

❑8.RECORD KEEPING

✓Objective: Record all information necessary to trace and determine the origin of pest or disease in the event of an outbreak.

❖Simple practices

➢Maintain records to trace stock, and their associated health status, onto, within or from the farm

➢Maintain records for all aspects of the biosecurity plan.

❑Other records that should be maintained include, but are not limited to:

➢all stock transfer on to, within, and off your farm

➢stock health, purchases and sales, monitoring and surveillance activities, testing and declarations, stocking densities, stock performance and feed schedules.

COMING UP WITH A BIOSECURITY PLAN

❑ 9.MONITORING/SURVEILLANCE

✓Objective: Minimise the biosecurity risk to a farm by implementing appropriate monitoring and surveillance practices.

❖Simple practices

➢Regularly inspect and monitor all of your stock and record abnormalities and mortalities (including early signs of stress, behavioural changes or presence of pests)

➢Investigate suspected health problems in your stock.

➢Observe and record abnormalities and mortalities in wild stocks around your farm.

➢Maintain access to qualified veterinarian or aquatic health professional who can provide advice and examine stock at short notice.

COMING UP WITH A BIOSECURITY PLAN

❑10. WASTE (OTHER THAN WATER)

✓Objective: Minimise the pest and disease risks associated with removal and disposal of farm waste.

❖Simple practices

- All waste should be assessed for biosecurity risk to the farm and environment and appropriate action(s) taken to address identified risks

➢Containment, handling and disposal of waste should be conducted in a biosecure manner

➢Designate waste holding areas away from production or feed storage areas

➢Contain, handle and dispose of waste in a biosecure manner

➢Disposal of waste should be to an approved waste site or waste management company.

COMING UP WITH A BIOSECURITY PLAN

❑11. AUDITING

✓Objective: To ensure your farm biosecurity plan continues to address biosecurity risks effectively and efficiently.

❖Simple practices

➢Have audits conducted of on-farm biosecurity plans and their implementation at regular prescribed intervals

➢Farms should have a veterinary health plan (VHP), which includes a biosecurity plan that sets out biosecurity protocols, preventive measures, treatments and contingencies.

➢Your farm biosecurity plan should include a schedule for routine review and identify any triggers for extraordinary review.

Fundamental for success

(Standard) operating procedures



Quarantine measures. (Isolation and observation)



Vehicles disinfection



Entering people, materials, and equipment to the production area



Animal health program (Including monitoring health Plan)



Side effects of veterinary supplies notification



Cleaning and disinfection for: facilities (cages, ponds, tanks), containers, storage rooms, equipment, etc.



Management and final disposal of sanitary risk materials and solid waste.



Movement of live animals



Unusual mortalities or alterations in the productive parameters notification



Management and final disposal of mortality



Importation certification or document of origin of the animals

SOME BENEFITS OF BIOSECURITY

- Improve market access
- Increase productivity
- Reduction in production costs
- Reduction rate of emergence antimicrobial resistance
- enhances food safety, protects the investments by preventing economic losses
- optimizing the health and immunity of fish stocks

AQUACULTURE BIOSECURITY IN ZAMBIA

Journal of Aquaculture Research & Development

OPEN ACCESS Freely available online

Review Article

Aquaculture Health Management Practices in Zambia: Status, Challenges and Proposed Biosecurity Measures

Oliver J. Hasimuna[1,2*], Sahya Maulu[3,4], Joseph Mphande[5]

CHALLENGES

- Lack of knowledge due to inadequate extension services
- Poor entry and exit control measures against fish and other organisms
- Lack of basic biosecurity measures
- Lack of proper diagnostic tools

PROPOSED BIOSECURITY MEASURES

There is need to manage animals, people, vehicles and other moving equipment, water and sediments, feed, and solid and liquid wastes

CONCLUSION

- Good biosecurity measures will reduce the risk of catastrophic losses from infectious disease and low-level losses that, over time, can also greatly affect the bottom line.
- Work with an experienced aquatic animal health professional, production specialist, and aquacultural engineer to develop a comprehensive biosecurity plan.
- Know your fish source and water source to ensure that specific pathogens don't enter your facility by either route.
- Quarantine, test, and continue to monitor fish health throughout the production cycle.
- Good husbandry is key to reducing disease susceptibility

CONCLUSION

- Regularly monitor systems, water quality, and fish. Isolate sick fish and remove dead and moribund fish.
- Know which diagnostics to use and which treatments are legal and available.
- Educate your personnel and visitors so that they understand and follow biosecurity protocols. Regularly evaluate educational programs.
- Plan your facility's sanitation, disinfection, and system management schemes and design your biosecurity with science, efficiency, and personnel compliance in mind.
- Keep good records. Compliance with, and documentation of, biosecurity protocols will help reduce losses during an outbreak and may help prevent mandatory depopulation.

EXAMPLE OF PLAN

Basic Biosecurity Plan Template

Use the following as a template for a general biosecurity plan. Additional information for recirculating aquaculture systems and pond systems will be covered in separate publications.

1. Which species is/are being cultured?
2. Draw a schematic of the farm. Include each building and each system, entry and exit points, and major flow patterns (fish movement, visitor and employee movement). Identify the life stages (eggs, juveniles, adults) found in each system.
3. System (pond, tank) management and husbandry
   - How will the system be managed to ensure good water quality/chemistry for the species?
   - Is the system appropriate for the given species, life stage, and density?
   - Can individual units be effectively isolated in the event of an outbreak?
   - Is cleaning the system easy?

   Changes should be made where possible to correct any problems.
4. What are the important infectious diseases for the species cultured?

   For example, diseases of concern for koi producers include koi herpesvirus (KHV) and spring viremia of carp (SVC); diseases of concern for tilapia producers include *Streptococcus* and *Francisella*.
5. Where can these disease-causing agents be found or how can they enter the farm?
   a. Water source: safe (biosecure) or unsafe (not biosecure)?
      - If safe (from a deep well, municipal supply, or originally from an unprotected source but disinfected before use), your water should pose little or no risk. **Note:** Water from a safe source is not always the best quality for the species being raised and may need to be treated.
      - If unsafe (from a shallow well or surface water), how will it be disinfected?

EXAMPLE OF PLAN

b. Fish:
   - Identify a limited number of well-known, reputable suppliers.
   - Have suppliers provided information on the health status of new fish?
   - Have their fish been tested for specific diseases?
   - How will you quarantine new fish and have them tested for specific pathogens of concern, if necessary?
   - How will you isolate sick fish from the rest of the population?

c. Diet/Nutrition:
   - Identify reputable sources for live and frozen foods.
   - Where will you store commercially prepared feeds?
   - Who is responsible for labeling and dating stored feeds?
   - When will feeds be inspected so that out-dated or possibly contaminated feeds can be discarded?

7. Management of disease outbreaks
   - What is the plan for observing fish daily for changes in feeding, behavior or appearance that might indicate a disease outbreak?
   - What steps will be taken if there is a disease outbreak?
     - Signs posted
     - Investigation (water quality, nutrition, disease diagnostics)
   - Are diagnostics and tests used to determine which legal drugs and chemicals to use for treatment?
   - Who will notify authorities for reportable diseases?
   - How will dead fish be disposed of to meet local, state and federal regulations?

EXAMPLE OF PLAN

6. How can diseases be spread? They can be spread by the fish, water, pests, other animals, food, people, and equipment.
   - How will the system be designed to separate different "lots" of fish?

     A "lot" of fish is a unit of animals of the same species, and usually the same life stage, that is managed as one group during production. Different lots of fish should be physically separated (different holding areas, ponds, systems, buildings), with fish of similar biosecurity status and age closest to each other and those of unknown or differing status (broodstock vs. fingerlings vs. eggs) most separate. Normally this means that the group has originated, been raised, and/or been housed together long enough (usually at least several weeks or months) to be considered of the same health and disease status and susceptibility.
   - Where should signs be posted to identify different security zones on the farm and to restrict the movement of employees and visitors?

7. Management of disease outbreaks
   - What is the plan for observing fish daily for changes in feeding, behavior or appearance that might indicate a disease outbreak?
   - What steps will be taken if there is a disease outbreak?
     - Signs posted
     - Investigation (water quality, nutrition, disease diagnostics)
   - Are diagnostics and tests used to determine which legal drugs and chemicals to use for treatment?
   - Who will notify authorities for reportable diseases?
   - How will dead fish be disposed of to meet local, state and federal regulations?
   - How will discharge from diseased systems be treated and/or disposed of to meet local, state and federal regulations?
   - How will the facilities be cleaned and disinfected to eliminate any pathogen reservoirs in holding areas, on equipment, or on any other surfaces?

EXAMPLE OF PLAN

8. Records

Records are important legal documents that can demonstrate facility operation and production as well as system isolation during reportable disease outbreaks (which may prevent unnecessary depopulation).

- Keep records of water quality parameters, changes of footbath and net dip solutions, feed expiration dates, animal inventory, fish disease outbreaks (including dates and number dead per day), disease investigation findings, and use of drugs and chemicals (including expiration dates and when used).

9. Biosecurity plan evaluation

The plan should be assessed at least once a year for effectiveness and compliance. Appropriate and up-to-date records will provide information about the increase or decrease in the number of losses, the use of drugs and chemicals, and sales.

Producers developing biosecurity plans should begin by examining their goals and evaluating the related risks. Biosecurity management should target the most important areas and pathogens/diseases to be protected against and implement strategies that are effective and practical. Compliance and documentation are the big challenges to a successful plan. Everyone working in or with a facility needs to have ownership in the plan and recognize why specific procedures are in place.